$R = \frac{\rho l}{A}$

EXPERIMENT NO. 31
ELECTRICAL CONDUCTIVITY AND BAND-GAP MEASUREMENT OF GERMANIUM FOUR PROBE METHOD

1. OBJECTIVES:

(i) To study the temperature dependence of the resistivity of germanium crystal.
(ii) To measure the band-gap of germanium.

2. INTRODUCTION:

According to the band theory of solids, insulators and semiconductors are materials which possess a band-gap (i.e., a range of forbidden energy values) at the Fermi level. Thus, these materials have a completely filled energy band below the gap and an empty band above the gap. The width of this band-gap is what distinguishes insulators from semiconductors. In semiconductors the band-gap is small enough so that at finite temperatures thermal excitation of electrons across the gap, into the empty "conduction" band, is possible leading to a small but measurable conductivity. The band-gap in insulators is simply too large to have any appreciable concentration of charge carriers excited into the conduction band. The temperature dependence of the resistivity of a pure (i.e., intrinsic) semiconductor is given by

$$\rho = B(T)\exp(Eg/2k_BT) = \rho_o e^{(Eg/2kBT)} \qquad \text{(i)}$$

where Eg is the width of the band-gap and the function $B(T)$ is only very weakly dependent on temperature. To a good approximation we can take $B(T)$=constant. Thus, we can easily measure the gap energy of a semiconductor material by measuring the resistivity/conductivity of a sample over a range of temperatures.

$$\sigma = \sigma_o \exp(-Eg / 2k_BT) \qquad \text{(ii)}$$

$$\ln \sigma = -(Eg / 2000k_B) \times (1000/T) + \ln \sigma_o \qquad \text{(iii)}$$

For $\ln \sigma$ vs. 1000/T curve: **Eg =slope x 0.172 eV**

OR

$$\log \sigma = (1/2.303)[-(Eg / 2000k_B) \times (1000/T)] + \log \sigma_o$$

For $\log \sigma$ vs. 1000/T curve: Eg =slope x 0.396 eV

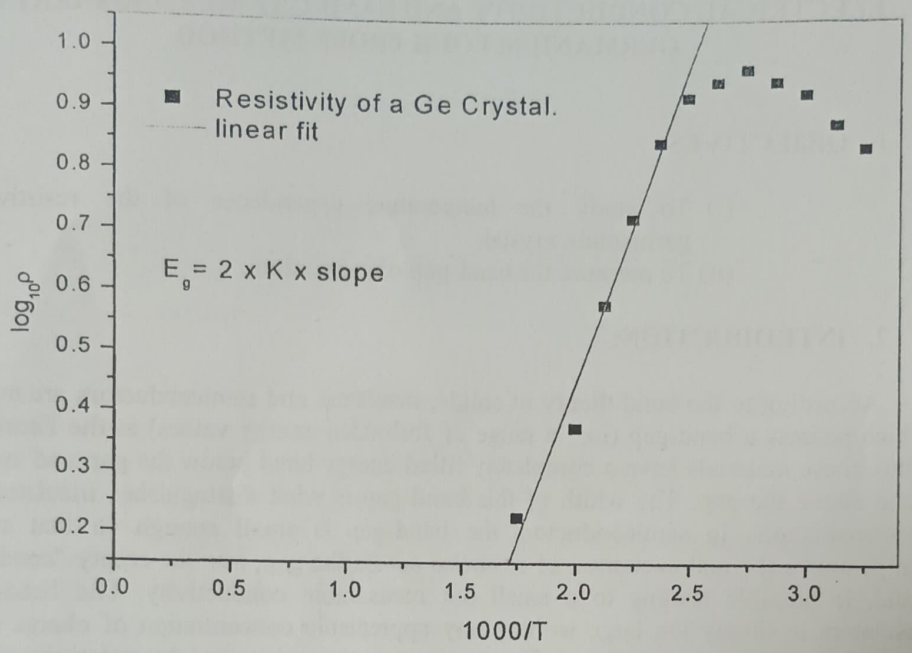


Fig.1: The resistivity of a Ge crystal as a function of inverse temperature. Eg can be calculated from slope of $\text{Log}_e\rho$ vs [1000/T] curve in intrinsic region.

EXPERIMENTAL PROCEDURES:

1. Connect the outer pair of probes (red/black) leads to the constant current power supply and the inner pair (yellow/green leads) to the probe voltage terminals.
2. Place the four probe arrangement in the oven and fix the thermocouple in the oven through the hole provided.
3. Switch on the ac mains of Four Probe Set-up and put the digital panel meter in the current measuring mode through the selector switch. In this position LED facing mA would glow. Adjust the current to a desired value (Say 5 mA).
5. Now put the digital panel meter in voltage measuring mode. In this position LED facing mV would glow and the meter would read the voltage between the probes.
6. Connect the oven power supply. Rate of heating may be selected with the help of a switch Low or High as desired. Switch on the power to the Oven. The glowing LED indicates the power to the oven is 'ON'. Note the values of thermo emf. And the corresponding value of the four probe voltage.
7. Take the reading for heating as well as cooling cycle. Repeat the experiment for different values of source current (say 10 mA and 15 mA).

OBSERVATION :

Current (I) = ... mA (Constant)
Distance between probes (S): ..... mm,
Thickness of the crystal (W): ..... mm.

S.No.	Temperature (^oC)	Voltage (volts)	Temperature (K)	ρ (ohm. cm.)	1000/T

CALCULATION:

ρ_o can be obtained from following equation:

$$\rho_o = (V/I) \times 2 \pi S$$

Since the thickness of the crystal is small compared to the probe distance, a correction factor for it has to be applied. Thus ρ may be calculated for various temperatures by

$$\rho(T) = \rho o / G7 \ (W/S)$$

(W/S) = 0.25 for this sample (see fig. 2 and table 1)

Plot a graph for $Log_{10} \rho$ vs. [1000/T]

$$\mathbf{Log_{10} \rho = [E_g/2k_BT]/2.3026 - log_{10} K}$$

the slope of the curve $Log_{10} \rho$ vs. [1000/T] is given= $(E_g/2000k_B)/2.3026$

E_g = slope x $2000k_B$ x 2.3026

$\mathbf{E_g}$ **= slope x 0.396 eV** (Value of Boltzmann's constant is taken in eV/K)

We can also calculate E_g from a graph for $Log_e \rho$ vs. [1000/T]

$$\mathbf{Log_e \rho = [E_g/2k_BT] - log_e K'}$$

$\mathbf{E_g}$ **= slope x 0.172 eV** (Value of Boltzmann's constant is taken in eV/K)

Thus Eg may be obtained from the slope of the graph. Note that **ln X** or $\mathbf{log_e X = 2.3026 \ log_{10} X}$ and above Eq. is applicable only in the intrinsic region of the semiconductor. A typical graph is shown in Fig. 1.

Table 1

S.No.	W/S	G6 (W/S)	G7 (W/S)
1	0.100	0.0000019	13.863
2	0.141	0.00018	9.704
3	0.200	0.00342	6.931
4	0.330	0.0604	4.159
5	0.500	0.2280	2.780
6	1.000	0.6830	1.504
7	1.414	0.8480	1.223
8	2.000	0.9330	1.094
9	3.333	0.9838	1.0228
10	5.000	0.9948	1.0070
11	10.000	0.9993	1.00045

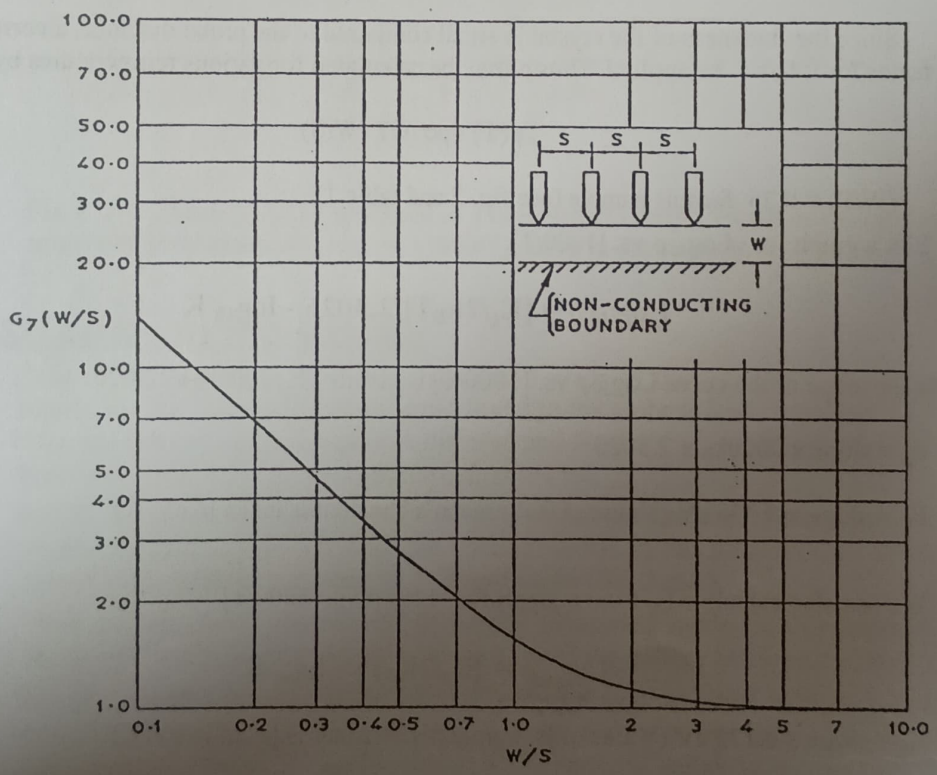


FIG.2 CORRECTION DIVISOR FOR PROBES ON A THIN SLICE WITH A NONCONDUCTING BOTTOM SURFACE

QUESTIONS:

1. What is the advantage of Four Probe method over the other conventional methods? Explain Ohmic and non-ohmic contacts.
2. Can we use an ordinary millivoltmeter instead of electronic millivoltmeter or potentiometer to measure the inner probe voltage? why?
3. Explain the behaviour of the log ρ vs. 1/T curve. Why does a semiconductor behaves as an insulator at zero degree Kelvin?
4. What are direct and indirect bandgap semiconductors. Give example of each.

References:

1. 'Introduction of Solid State Physics', by C. Kittel 1Vth edition.
2. 'Fundamental of Semiconductor Devices', by J. Lindmayer and C.Y. Wriegley, 1071 East West Press.
3. 'Hand Book of Semiconductor Electronics', by L.P. Hunter (ed.) MeGraw Hill Book Co, Inc. NY (1962).
4. 'Experiments in Modern Physics', by A.C. Melissions, Academic Press, NY (1966).
5. "Resistivity of semiconductors by four probe method at different temperatures" Users Manual, Scientific Equipment and services.

USEFUL CONSTANTS: Four Probe sample: Ge crystal
$\rho \approx$ 6-8 Ωcm
Thickness(w)=0.50mm
Distance between probes=0.200cm
Standard value of Band Gap of Ge=0.67eV